

# Generation Interconnection System Impact Study Report

## For Queue Project Q20-1445

Pheasant Branch 138 kV

August 28, 2021

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1445. The Interconnection Customer proposes a 721.6 MW natural gas combined cycle facility interconnecting at the Pheasant Branch 138 kV substation in the Virginia area. The request was received on April 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 480 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,880,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

For the loss of Pheasant Branch to Imperial Savannah Lp 138 kV, the Pheasant Branch Substation to Switzer 138 kV loads to 416.7% of its 165 MVA rating. Project DFAX: 100.0%.

Imperial Savannah Lp to Pheasant Branch 138 kV line: 327.2% loading (210 MVA) following the outage of the Switzer to Pheasant Branch 138 kV; DFAX = 100.0%.

The Switzer to Plaza 138 kV is overloaded to 318.6% of its 205 MVA emergency rating for the loss of the Imperial Savannah Lp to Pheasant Branch 138 kV. The project contributes 100.0% DFAX (721.6 MW) to this overload.

The Imperial Savannah Lp to Haggin 138 kV line is overloaded to 306.3% of its 220 MVA emergency rating for the loss of the Switzer to Pheasant Branch 138 kV. The project contributes 100.0% DFAX (721.6 MW) to this overload.

The Plaza to Big Hanaford 138 kV is overloaded to 139.3% of its 287 MVA emergency rating for the loss of the Mitchell Dam to Skaggs Island Junction 2 345 kV Line. The project contributes 31.7% DFAX (228.7 MW) to this overload.

Fritch to Lakeside 138 kV Line: 112.4% loading (303 MVA) following the outage of the Skaggs Island Junction 2 to Mitchell Dam 345 kV; DFAX = 18.9%.

For the loss of Plaza to Big Hanaford 138 kV, the Carver to Big Hanaford 138 kV loads to 106.7% of its 215 MVA rating. Project DFAX: 22.2%.

For the loss of Plaza to Big Hanaford 138 kV, the Carver to Shockoe 138 kV loads to 103.6% of its 289 MVA rating. Project DFAX: 22.2%.

For the loss of Leavitt to Birchwood 345 kV, the Trenton Diesel to Four Rivers 138 kV loads to 102.9% of its 290 MVA rating. Project DFAX: 5.2%.

For the loss of Mitchell Dam to Haggin 345 kV, the Turner Rd to Tyler 138 kV loads to 101.8% of its 335 MVA rating. Project DFAX: 11.0%.

For the loss of Switzer to Pheasant Branch 138 kV, the Haggin to Tyler 138 kV loads to 100.2% of its 393 MVA rating. Project DFAX: 29.2%.

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Pheasant Branch to Switzer 138 kV                   | Rebuild line with higher-capacity conductor | \$1,704,000   | 100% (\$1,704,000)    |
| Imperial Savannah Lp to Pheasant Branch 138 kV line | Rebuild line with higher-capacity conductor | \$1,736,000   | 100% (\$1,736,000)    |
| Switzer to Plaza 138 kV                             | Rebuild line with higher-capacity conductor | \$1,670,000   | 100% (\$1,670,000)    |
| Haggin to Imperial Savannah Lp 138 kV               | Rebuild line with higher-capacity conductor | \$13,122,000  | 100% (\$13,122,000)   |
| Plaza to Big Hanaford 138 kV                        | Rebuild line with higher-capacity conductor | \$6,807,000   | 56% (\$3,843,000)     |
| Fritch to Lakeside 138 kV Line                      | Rebuild line with higher-capacity conductor | \$2,805,000   | 32% (\$892,000)       |
| Carver to Big Hanaford 138 kV                       | Reconductor line                            | \$996,000     | 57% (\$573,000)       |
| Carver to Shockoe 138 kV                            | Reconductor line                            | \$341,000     | 57% (\$196,000)       |
| Trenton Diesel to Four Rivers 138 kV                | Reconductor line                            | \$2,764,000   | 31% (\$856,000)       |
| Tyler Substation to Turner Rd 138 kV                | Reconductor line                            | \$3,176,000   | 36% (\$1,141,000)     |
| Haggin Substation to Tyler 138 kV                   | Reconductor line                            | \$231,000     | 64% (\$147,000)       |
|                                                     | Total allocated to project                  |               | \$25,880,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.